North Ridge Highway Intersection Upgrade

Existing conditions and ground summary

1. Road and survey basis

The existing paved surface is 12.0 m wide, with two 3.5 m painted through lanes and two 2.5 m paved shoulders. The north industrial access forms a T-intersection near chainage 0+300. Local working coordinates and elevations are used in the review drawings.

Preliminary measured eastbound stopping sight distance at the crest is 165 m. This value requires validation against the adopted eye/object heights, actual grades and the final accepted design basis.

2. Profile and sight-distance review

Vertical curve: BVC 0+175 at 670.000 m; incoming grade +3.0%; outgoing grade -3.0%; curve length 250 m. PVI is at 0+300 with tangent-intersection elevation 673.750 m. The road-surface crest at 0+300 is 671.875 m.

Calculated K = 250 / 6 = 41.67 m/%. The current profile is retained for review only. A grade-corrected sight-distance check and any exception submission remain outstanding.

Project	Issue information
NRH-26104	2026-08-28
North Ridge Industrial Lands Partnership	Issued for review
Prepared / checked	JL / AK

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3. Traffic and ground inputs

2026 base-year AADT is 7,800 vehicles/day with 14% heavy vehicles. The traffic assessment adopts 2.0% annual non-compounded background growth. Truck classification and axle-load spectra require confirmation for pavement design.

Preliminary subgrade design input is CBR 5%. Localized cracking and edge deterioration are noted. Coring, laboratory testing, drainage assessment and confirmation of seasonal conditions are required before final pavement design.